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Ring statistics analysis of topological networks: New approach and application to amorphous GeS₂ and SiO₂ systems

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ABSTRACT

In a first part we present the state of the art of the ring statistics when used to study the structural properties of solids. The different possibilities used to classify rings when analysing a material using the graph theory framework are presented. Afterwards a new and original method is proposed to analyse the connectivity of a topological network using ring statistics. This method allows to obtain the connectivity profile of the network and then to study and compare with efficiency the connectivity of different networks using ring statistics. Then a detailed analysis of SiO₂ and GeS₂ amorphous samples obtained using molecular dynamics is proposed to illustrate the main features of our method. A FORTRAN 90 code mainly devoted to the ring statistics analysis of materials with an implementation of the present method is also presented.

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1. Introduction

Since its introduction in 1967 by King [1] the analysis of the connectivity of topological networks via the ring statistics has been developed and improved in a significant way. Presently several definitions of rings exist [2–6] and can be used to study a material using these techniques. When studying amorphous systems ring statistics is mainly used to obtain information inaccessible using standard tools such as the radial distribution functions and the neutron or X-rays structure factors. In particular it allows to scan the network at a farther scale than the first or the second coordination shell. Thus it is possible to obtain information on the Intermediate Range Order 'IRO' of the glass.

It is well known that in glass science finding the origin of the disorder and understanding the origin of the IRO is a key for the interpretation of the optical and conduction phenomena. In this context using the ring statistics techniques appears to be fully justified. Nevertheless precise information can not be obtained without using an accurate method.

In the first part of this article we present a brief review of the different possibilities used to classify rings when analysing a material using the graph theory framework. In a second part we propose

an original method to analyse the connectivity of a topological network using ring statistics. This method allows to obtain the connectivity profile of the network and then to study and compare with efficiency the connectivity of different networks using ring statistics. In a third and last part a detailed study of amorphous SiO₂ and GeS₂ samples obtained using molecular dynamics simulations is proposed to illustrate the main features of our method.

An appendix of the present article is dedicated to the presentation of a code developed in FORTRAN 90 to implement this method as well as other tools useful to analyse the results of molecular dynamics simulations.

2. Ring statistics: state of the art

The analysis of topological networks (liquid, crystalline or amorphous systems) is often based on the part of the structural information which can be represented in the graph theory using nodes for the atoms and links for the bonds. The absence or the existence of a link between two nodes is determined by the analysis of the total and partial radial distribution functions of the system.

In such a network a series of nodes and links connected sequentially without overlap is called a path. Following this definition a ring is therefore simply a closed path. If we study thoroughly a specific node of this network we see that this node can be involved

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in numerous rings. Each of these rings is characterized by its size and can be classified based upon the relations between the nodes and the links which constitute it.

2.1. Size of the rings

There are two possibilities for the numbering of rings. On the one hand, one can use the total number of nodes of the ring, therefore a N -membered ring is a ring containing N nodes. On the other hand, one can use the number of *network forming nodes* (ex: Si atoms in SiO_2 and Ge atoms in GeS_2 which are the atoms of highest coordination in these materials) an N -membered ring is therefore a ring containing $2 \times N$ nodes. For crystals and SiO_2 -like glasses the second and mostly used method is satisfactory. Nevertheless the first method has to be used in the case of chalcogenide liquids and glasses in order to count rings with homopolar bonds (ex: Ge–Ge and S–S bonds in GeS_2) – see Section 4.1 for further details.

The ring statistics presented in the following study have been carried out using 30 nodes as maximum search depth, i.e. as maximum size for a ring. This value has been chosen to get the best possible compromise between CPU time and quality of the description. The CPU time indeed increases exponentially with the maximum search depth [Appendix C]. Furthermore in the networks we have studied thereafter there are only very few larger rings. However, from a theoretical point of view it is possible to obtain an idea of the maximum size a ring could have in the network. This theoretical maximum size will depend on the properties of the disordered system studied as well as on the criterion used to define a ring.

2.2. Definitions

2.2.1. The King's criterion

The first way to define a ring has been given by King [1]. In order to study the connectivity of glassy SiO_2 she defines a ring as the shortest path which comes back to a given node (or atom) from one of its nearest neighbors [Fig. 1].

In the case of the King's criterion one can calculate the maximum number of different ring sizes, $NS_{\max}(\text{King})$, which can be found using the atom At to initiate the search:

$$NS_{\max}(\text{King}) = N_c(\text{At}) - 1 \quad (1)$$

where $N_c(\text{At})$ is the number of neighbors of atom At . $NS_{\max}(\text{King})$ represents the number of ring sizes found if the neighbors of atom At are connected together with paths of different sizes.

It is also possible to calculate the Theoretical Maximum Size, $TMS(\text{King})$, of a King's ring in the network using:

$$TMS(\text{King}) = 2 \times D_{\max} \quad (2)$$

where $D_{\max}$ represents the longest distance, in terms of chemical bonds, separating two atoms in the network. If used when looking for rings, periodic boundary conditions have to be taken into account to calculate $D_{\max}$. The relation [Eq. (2)] is illustrated in Fig. 4-1.

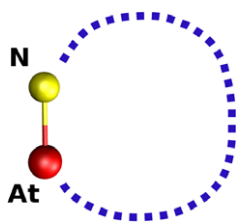


Fig. 1. King's criterion in the ring statistics: a ring represents the shortest path which comes back to a given node (At) from one of its nearest neighbors (N).

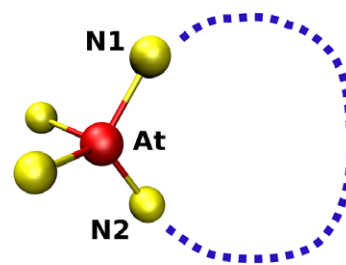


Fig. 2. Shortest path criterion in the ring statistics: a ring represents the shortest path between two of the nearest neighbors ($N1$ and $N2$) of a given node (At).

2.2.2. The shortest paths criterion

Using the King's criterion, in some cases, shortest paths were ignored by the algorithm. With the shortest path criterion [2,3], the algorithm has been modified and a ring represents the shortest path between two of the nearest neighbors of a given node [Fig. 2].

Differences between the King and the shortest path criteria are illustrated in Fig. 3.

Like for the King's criterion, with the shortest path criterion one can calculate the maximum number of different ring sizes, $NS_{\max}(\text{SP})$, which can be found using the atom At to initiate the search:

$$NS_{\max}(\text{SP}) = \frac{N_c(\text{At}) \times (N_c(\text{At}) - 1)}{2} \quad (3)$$

where $N_c(\text{At})$ is the number of neighbors of atom At . $NS_{\max}(\text{SP})$ represents the number of ring sizes found if all couples of neighbors of atom At are connected together with paths of different sizes.

It is also possible to calculate the theoretical maximum size, $TMS(\text{SP})$, of a shortest path ring in the network using:

$$TMS(\text{SP}) = 2 \times (N_{c_{\max}} - 1) \times (D_{\max} - 1) \quad (4)$$

where $D_{\max}$ is the longest distance, in terms of chemical bonds, separating two atoms in the network, and $N_{c_{\max}}$ represents the average number of neighbors of the chemical species of highest coordination. If used when looking for rings, periodic boundary conditions have to be taken into account to calculate $D_{\max}$. The relation [Eq. (4)] is illustrated in Fig. 4-2.

Since the introduction of the King's and the shortest path criteria other definitions of rings have been proposed. These definitions are based on the decomposition of rings into smaller rings.

2.2.3. The primitive rings criterion

A ring is primitive [4,5] (or irreducible [6]) if it can not be decomposed into two smaller rings [Fig. 5].

The primitive rings analysis between the paths in Fig. 5 may lead to three results depending on the relations between the paths A, B, and C:

- If paths A, B, and C have the same length: $A = B = C$ then the rings 'AB', 'AC' and 'BC' are primitive rings.
- If the relation between the paths is like $? = ? < ?$ (ex: $A = B < C$) then 1 smaller ring ('AB') and two bigger rings ('AC' and 'BC') exist. None of these rings can be decomposed into the sum of two smaller rings therefore the three rings are again primitive rings.
- If the relation between the path is like $? < ? = ?$ (ex: $A < B = C$) or $? < ? < ?$ (ex: $A < B < C$) then a shortest path exists (A). It will be possible to decompose the ring ('BC') built without this shortest path into the sum of two smaller rings ('AB' and 'AC'), therefore this ring will not be a primitive ring.

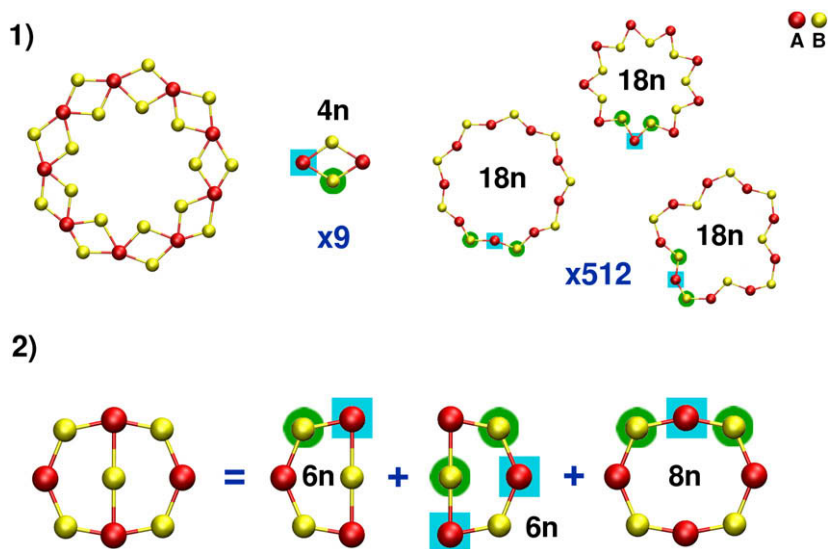


Fig. 3. Differences between the King and the shortest path criteria for the ring statistics in an AB_2 system. In these two examples the search is initiated from chemical species A (blue square). The nearest neighbor(s) of chemical species B (green circles) are used to continue the analysis. (1) In the first example only rings with four nodes are found using the King's criterion, whereas rings with 18 nodes are also found using the shortest path criterion (2^9 rings with 18 nodes). (2) In the second example the shortest path criterion allows to find the ring with eight nodes ignored by the King's criterion which is only able to find the rings with six nodes. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

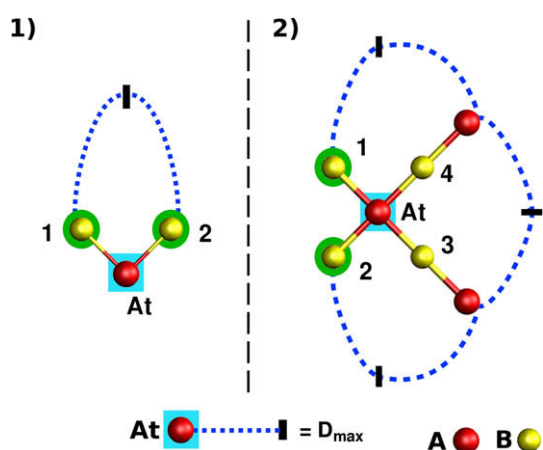


Fig. 4. Theoretical maximum size of the rings for an AB_2 system ($N_{C_{max}} = N_{C_A} = 4$) and using: (1) the King's criterion, (2) the shortest path criterion. The theoretical maximum size represent the longest distance between two nearest neighbors 1 and 2 (green circles) of the atom At used to initiate the search (blue square). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

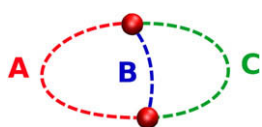


Fig. 5. Primitive rings in the ring statistics: the 'AC' ring defined by the sum of the A and the C paths is primitive only if there is no B path shorter than A and shorter than C which allows to decompose the 'AC' ring into two smaller rings 'AB' and 'AC'.

2.2.4. The strong rings criterion

The strong rings [4,5] are defined by extending the definition of primitive rings. A ring is strong if it can not be decomposed into a sum of smaller rings whatever this sum is, ie. whatever the number of paths in the decomposition is.

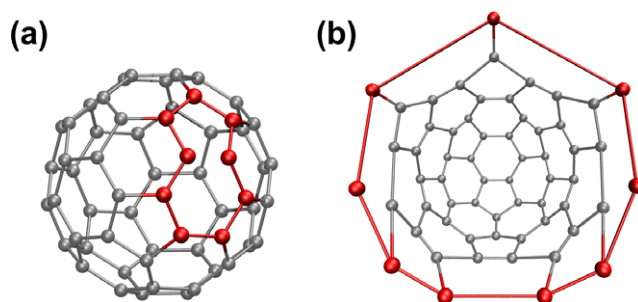


Fig. 6. Strong rings in the ring statistics: (a) the 9-carbon-atoms ring created after breaking a C-C bond in a Buckminster fullerene molecule is a counterexample of strong ring; (b) the combination of shortest rings, 11 5-carbon-atoms rings and 19 6-carbon-atoms rings, appears easily after the deformation of the C_{60} molecule.

By definition the strong rings are also primitives, therefore to search for strong rings can be reduced to find the strong rings among the primitive rings. This technique is limited to relatively simple cases, like crystals or structures such as the one illustrated in Fig. 6. On the one hand the CPU time needed to complete such an analysis for amorphous systems is very important. On the other hand it is not possible to search for strong rings using the same search depth than for other types of rings. The strong ring analysis is indeed diverging which makes it very complex to implement for amorphous materials.

In the case of primitive rings like in the case of strong rings, there is no theoretical maximum size of rings in the network.

3. Description of a network using ring statistics

3.1. Existing tools

Rings statistics is mainly used to obtain a snapshot of the connectivity of a network. Thereby the better the snapshot will be, the better the description and the understanding of the properties of the material will be. In the literature many articles present studies of materials using ring statistics. In these studies either the

number of Rings per Node ' R_N ' [7,8] or the number of Rings per Cell ' R_C ' [9–11] are given as a result of the analysis. The first (R_N) is calculated for one node by counting all the rings corresponding to the property we are looking for (King's, shortest path, primitive or strong ring criterion). The second (R_C) is calculated by counting all the different rings corresponding at least once (at least for one node) to the property we are looking for (King's, shortest path, primitive or strong ring criterion). The values of R_N and R_C are often reduced to the number of nodes of the networks. Furthermore the results are presented according to each size of rings.

An example is proposed with a very simple network illustrated in Fig. 7a. This network is composed of 10 nodes, arbitrary of the same chemical species, and seven bonds. Furthermore it is clear that in this network there are one ring with three nodes and one ring with four nodes. It is easy to calculate R_N and R_C for the network in Fig. 7a (n – number of nodes):

In the literature the values of R_N and R_C are usually given separately [7–11]. Nevertheless these two properties are not sufficient in order to describe a network using rings.

A simple example is proposed in Fig. 7, the two networks [Fig. 7a and b] do have very similar compositions with 10 nodes and seven links but they are clearly different. Nevertheless the previous definitions of rings per cell and rings per node even taken together will lead to the same description for these two different networks [Table 1].

In both cases (a) and (b) there are one ring with three nodes and one ring with four nodes. It has to be noticed that these two rings have properties which correspond to each of the definitions introduced previously (King, shortest path, primitive and strong). Thus none of these definitions is able to help to distinguish between these two networks. Therefore even though these simple networks are different, the previous definitions lead to the same description.

Thereby it is justified to wonder about the interpretation of the data presented in the literature for amorphous systems with a much higher complexity.

3.2. Rings and connectivity: a new approach

The first goal of ring statistics is to give a faithful description of the connectivity of a network and to allow to compare this information with others obtained for already existing structures. It is therefore important to find a guideline which allows to establish a distinction and then a comparison between networks studied using ring statistics. In the present paper we propose a new method to achieve this goal. First of all we noticed fundamental points that must be considered to get a reliable and transferable method:

1. The results must be reduced to the total number of nodes in the network. The nature of the nodes used to initiate the analysis when looking for rings will have a significant influence, therefore it is essential to reduce the results to a value for one node. Otherwise it would be impossible to compare the results to the ones obtained for systems made of nodes (particles) of different number and/or nature.
2. Different networks must be distinguishable whatever the method used to define a ring. Indeed it is essential for the result of the analysis to be trustworthy independently of the method used to define a ring (King, shortest paths, primitives, strong). Furthermore this will allow to compare the results of these different ring statistics.

3.2.1. Number of rings per cell ' R_C '

We have already introduced this value, which is the first and the easiest way to compare networks using ring statistics.

In the most simple cases, such as the one represented in Fig. 8, the networks can be distinguished using only the number of rings [Table 2]. Nevertheless in most of the cases more information is required to describe accurately the connectivity of the networks.

3.2.2. Description of the connectivity: difference between rings and nodes

The second information needed to investigate the properties of a network using rings is the evaluation of the connectivity between rings. Indeed the distribution of the ring sizes gives a first information on the connectivity, nevertheless it can not be exactly evaluated unless one studies how the rings are connected. The impact of the relations between rings, already presented in Fig. 7, has been illustrated in detail in Fig. 9. Fig. 9 represents the different possibilities to combine two rings with six nodes and one ring with four nodes in a network composed of 16 nodes.

It has to be noticed that for convenience several tables which illustrate the results of the following discussion have been moved to Appendix A.

Among the nine networks presented in Fig. 9 none can be distinguished using the R_C value [Table 3].

Furthermore it is not possible to distinguish these networks using the R_N value. It seems possible to isolate the case (a) [Table 4] from the other cases (b) → (i) [Table 4]. Nevertheless the results obtained using the primitive rings criterion are similar for all cases (a) → (i) [Table 4], this is in contradiction with the second statement [2] proposed in our method.

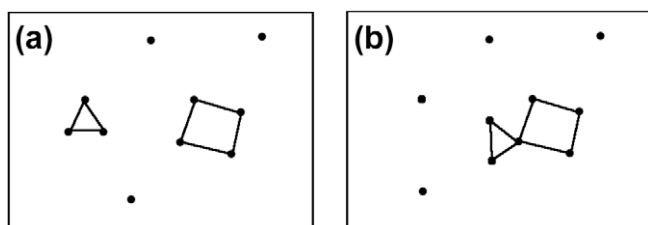


Fig. 7. Two simple networks having very close compositions: 10 nodes and seven links.

Table 1
 R_N and R_C calculated for the networks illustrated in Fig. 7.

n	$R_C(n)$	$R_N(n)$
3	1/10	3/10
4	1/10	4/10

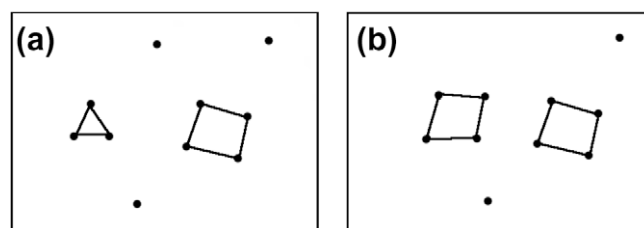


Fig. 8. The first comparison element: the total number of rings in the network.

Table 2
Number of rings in the simple networks represented in Fig. 8.

n	$R_C(n)$	
	(a)	(b)
3	1/10	0/10
4	1/10	2/10

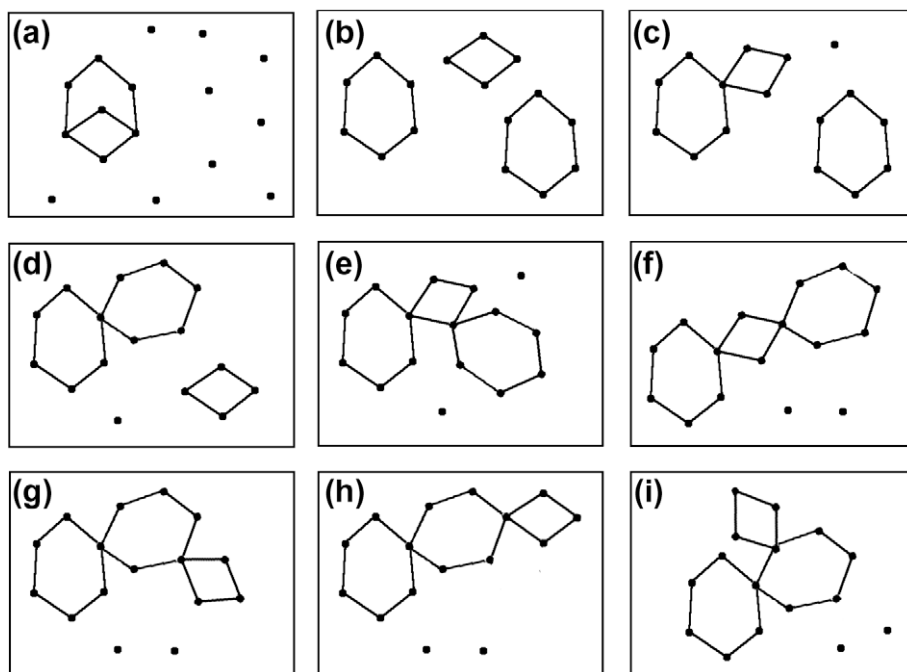


Fig. 9. Illustration of the nine different networks with 16 nodes, composed of two rings with six nodes and one ring with four nodes.

Table 3
Number of rings for the different networks presented in Fig. 9.

n	$R_C(n)$
4	1/16
6	2/16

Table 4
Number of rings per node for the networks presented in Fig. 9.

n	$R_N(n)$	
	King/S.P.	Primitive/strong
Case (a)		
4	4/16	4/16
6	10/16	12/16
n	$R_N(n)$	
	All criteria	
Cases (b) $\rightarrow$ (i)		
4	4/16	
6	12/16	

Before introducing parameters able to distinguish the configurations presented in Fig. 9 it is important to wonder about the number of cases to distinguish. From the point of view of the connectivity of the rings, configurations (a), (b), (c) and (d) are clearly different. Nevertheless following the same approach configurations (e) and (f) on the one hand and configurations (g), (h) and (i) on the other hand are identical. A schematic representation [Fig. 10] is sufficient to illustrate the similarity of the relations between these networks. The difference between each of these networks does not appear in the connectivity of the rings but in the connectivity of the particles. Thus if we analyse these models considering the ring as the base unit of the network, then these rings have an identical connectivity in the cases (e) and (f), and (g), (h) and (i) respectively. In the networks (e) and (f) for example two rings of size 6 are connected to one ring of size 4, and to find the difference between these two networks it is mandatory to study the connectivity of each particle.

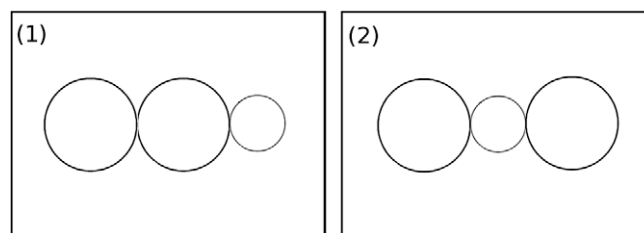


Fig. 10. Schematic representation of cases (g) $\rightarrow$ (i) (1) and (e) $\rightarrow$ (f) (2) illustrated in Fig. 9.

Table 5
General connectivity matrix.

$C_{mat} =$	$\begin{bmatrix} P_N(r) & P_N(r+1, r) & \cdots & P_N(R, r) \\ P_N(r, r+1) & \ddots & & P_N(R, r+1) \\ \vdots & & \ddots & \vdots \\ P_N(r, R) & \cdots & \cdots & P_N(R) \end{bmatrix}$
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Therefore among the networks illustrated in Fig. 9 six dispositions of the rings have to be distinguished (a, b, c, d, e, g). The proportions of particles involved, or not involved, in the construction of rings will become an important question.

The new tool defined in our method is able to describe accurately the information still missing on the connectivity. It is a square symmetric matrix of size $(R - r) \times (R - r)$, where R and r represent respectively the biggest and the smallest size of a ring found when analyzing the network: we have called this matrix the connectivity matrix [Table 5].

The diagonal elements $P_N(i)$ of this matrix represent the Proportion of Nodes at the origin of at least one ring of size i . And the non-diagonal elements $P_N(i, j)$ represent the Proportion of Nodes at the origin of ring(s) of size i and j .

The matrix elements have a value ranging between 0 and 1. The lowest and non equal to 0 is of the form $\frac{1}{Nn}$, the highest and non

equal to 1 is of the form $\frac{Nn-1}{Nn}$, where Nn represents the number of nodes in the network.

The connectivity matrix of the configurations illustrated in Fig. 9 allows to distinguish each network whatever the way used to define a ring is [Table A.6]. This matrix remains simple for small systems (crystalline or amorphous) or when using a small maximum ring size for the analysis. Nevertheless its reading can be considerably altered when analysing amorphous systems with a high maximum ring size for the analysis.

To simplify the reading and the interpretation of the data contained in this matrix for more complex systems, we chose a similar approach to extract information on the connectivity between the rings. As a first step we decided to evaluate only the diagonal elements $P_N(n)$ of the general connectivity matrix. Indeed these values allow us to obtain a better view of the connectivity than the standard R_N value.

Using $P_N(n)$ clearly improves the separation between the networks illustrated in Fig. 9. Nevertheless $P_N(n)$ does not allow to distinguish each of them yet [Table A.7]. We notice that the distinction between networks is improved [Table A.7] in particular when compared to the one obtain with $R_N(n)$ [Table 4].

Therefore in a second step we chose to calculate two properties whose definitions are very similar to the one of $P_N(n)$. The first, named $P_{N_{\max}}(n)$, represents the proportion of nodes for which the rings with n nodes are the longest closed paths found using these nodes to initiate the search. The second named, $P_{N_{\min}}(n)$, represents the proportion of nodes for which the rings with n nodes are the shortest closed paths found using these nodes to initiate the search.

The terms *longest* and *shortest path* must be considered carefully to avoid any confusion with the terms used in Section 2.2 to define the rings. For one node it is possible to find several rings whose properties correspond to the definitions proposed previously (King's, shortest path, primitive or strong ring criterion). These rings are solutions found when looking for rings using this particular node to initiate the analysis. In order to calculate $P_{N_{\max}}(n)$ and $P_{N_{\min}}(n)$ the longest and the shortest path have to be determined among these different solutions.

$P_{N_{\max}}(n)$ and $P_{N_{\min}}(n)$ have values ranging between 0 and $P_N(n)$. The lowest and non equal to 0 is of the form $\frac{1}{Nn}$, the highest and non equal to 1 is of the form $\frac{Nn-1}{Nn}$, where Nn represents the total number of nodes in the network. For the minimum ring size, $s_{\min}$, existing in the network or found during the search, $P_{N_{\min}}(s_{\min}) = P_N(s_{\min})$. In the same way for the maximum ring size, $s_{\max}$, existing in the network or found during the search, $P_{N_{\max}}(s_{\max}) = P_N(s_{\max})$.

To clarify this information it is possible to normalize $P_{N_{\max}}(n)$ and $P_{N_{\min}}(n)$ by $P_N(n)$. By reducing these values we obtain, for each size of rings, values independent of the total number of nodes Nn of the system. Then for a considered ring size the values only refer to the number of nodes where the search returns rings of this size:

$$P_{\max}(n) = \frac{P_{N_{\max}}(n)}{P_N(n)} \quad \text{and} \quad P_{\min}(n) = \frac{P_{N_{\min}}(n)}{P_N(n)} \quad (5)$$

The normalized terms $P_{\max}(n)$ and $P_{\min}(n)$ have values ranging between 0 and 1. The lowest and non equal to 0 is of the form $\frac{1}{Nn}$, the highest and non equal to 1 is of the form $\frac{Nn-1}{Nn}$. For the minimum ring size, $s_{\min}$, existing in the network or found during the search, $P_{\min}(s_{\min}) = 1$. In the same way for the maximum ring size, $s_{\max}$, existing in the network or found during the search, $P_{\max}(s_{\max}) = 1$.

$P_{\max}(n)$ and $P_{\min}(n)$ give complementary information to the ones obtained with $R_C(n)$ and $P_N(n)$ in order to distinguish and compare networks using ring statistics. The illustration of this re-

sult and therefore the complete information obtained with this method for the networks represented in Fig. 9 can be found in Table A.8.

$P_{\max}(n)$ and $P_{\min}(n)$ give information about the connectivity of the rings with each other as a function of their size. If a ring of size n is found using a particular node to initiate the search, $P_{\max}(n)$ gives the probability that this ring is the longest ring which can be found using this node to initiate the search. At the opposite, $P_{\min}(n)$ gives the probability that this ring is the shortest ring which can be found using this node to initiate the search.

Thereafter we will use the terms 'connectivity profile' to designate the results of a ring statistics analysis. This profile is related to the definition of rings used in the search and is made of the four values defined in our method: $R_C(n)$, $P_N(n)$, $P_{\max}(n)$ and $P_{\min}(n)$.

4. Application to AB₂ glasses

4.1. Introduction

Ring statistics have been developed mainly to study the connectivity of amorphous materials. However, the wide variety of sequences and structural defects that may be encountered in these materials make this type of analysis particularly challenging.

Some scenarios should therefore be reviewed, to lay the groundwork to look for rings in these compounds. In particular we need to evaluate the influence of different bond defects in the amorphous network. However, in each case both the impact on the connectivity, and the impact on the physical properties which can be deduced from the analysis, have to be studied. Furthermore it is necessary to determine if the influence of the bond defects depends on the criteria used to define a ring [Section 2.2].

It is clear that in-depth studies are necessary to fully investigate the influence of bond defects. Nevertheless to shorten analysis and discussions the next part of the article will be only dedicated to the impact on the connectivity studied using the shortest path criterion.

4.1.1. ABAB and BABA rings

The ring statistics of amorphous networks are often focused on finding rings made of a succession of atoms with an alternation of chemical species, called ABAB rings. The most common examples come from the alternation of Si and O atoms (in silica polymorphs) or Ge and S (in GeS₂ polymorphs). These solids are usually built with tetrahedra (SiO₄ or GeS₄) therefore we study the network distribution of tetrahedra.

The ideal technique to setup the analysis of such systems is to choose the atoms of highest coordination to initiate the search, respectively Si in SiO₂ and Ge in GeS₂. In most cases all rings can be found using this method. Nevertheless we can demonstrate that some solutions, so some rings, can be ignored by this analysis. This is highlighted in Fig. 11 which represents a cluster of atoms isolated from an AB₂ amorphous network.

We can see that this piece of network is characterized by a bond defect. An atom of the B species appears to be over-coordinated by three atoms of the A species. When looking for rings, using the shortest path criterion [Section 2.2.2] and initiating the search using the A atoms, the central ring with 10 nodes is ignored. Nevertheless other rings with 10 nodes are found and stored as solutions of the analysis. In order to find the central ring the search has to be initiated from the over-coordinated B atom.

By analogy with the terminology ABAB this ring can be called a BABA ring. Indeed the alternation of chemical species is well respected. Therefore it is legitimate to question the relevance of the analysis without this result. In other words we have to check out if this BABA ring is, or not, an ABAB ring.

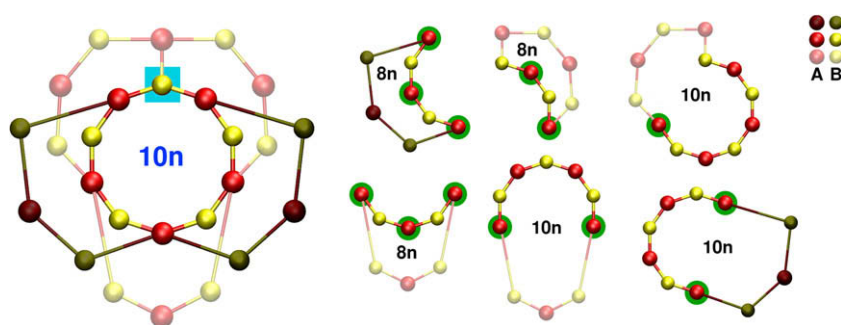


Fig. 11. Cluster of atoms isolated from an AB_2 amorphous network. A bond defect is located on an atom of the chemical species B (blue square). When looking for shortest paths [Section 2.2.2] using the chemical species A to initiate the search the central ring with 10 nodes is ignored. However among the solutions of the analysis (with the initial nodes circled in green) other rings with 10 nodes are found in the network. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

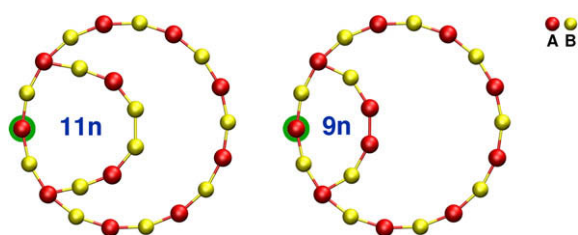


Fig. 12. Illustration of the influence of homopolar bonds in ABAB rings: in both examples the smallest rings found when initiating the search using the circled nodes (green color) contain an homopolar bond A–A or B–B. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The properties of this ring meet the definition and can therefore improve the description of the connectivity of the network. This kind of coordination defect [Fig. 11] is uncommon in vitreous silica [7,12], nevertheless it is frequent in chalcogenide glasses [13,14]. Therefore it is justified to discuss the impact of such BABA rings in the analysis of these compounds.

4.1.2. Homopolar bonds

In amorphous materials the homopolar bond defects can have a significant influence on the ring statistics. This is true in particular for AB_2 chalcogenide glasses. Fig. 12 illustrates standard cases that may be encountered when looking for rings in an AB_2 system which contains homopolar bonds.

The smallest rings found when initiating the search using the circled nodes (green color) are not ABAB rings. Therefore their size must be given using the total number of nodes. In Fig. 12 the smallest rings are a ring with nine nodes and a ring with 11 nodes containing respectively an A–A and a B–B homopolar bond. These rings are significantly smaller than the shortest ABAB ring with 18 nodes that may be found when looking for rings using the same green-circled nodes to initiate the analysis [Fig. 12]. Therefore it is once again justified to discuss the impact of such bond defects when analysing the connectivity of amorphous chalcogenides using ring statistics.

4.2. Ring statistics of SiO_2 and GeS_2

4.2.1. Theoretical framework

The connectivity of glassy $g-SiO_2$ [15] and $g-GeS_2$ [14] samples obtained using molecular dynamics simulations has been studied. These glasses are composed of 648 and 258 particles for respectively the $g-SiO_2$ ($\rho = 2.17 \text{ g/cm}^3$) and the $g-GeS_2$ ($\rho = 2.75 \text{ g/cm}^3$) sample. In both cases a similar framework has been used. The anal-

ysis has been performed using 30 nodes as maximum search depth and the results have been averaged over 100 configurations. Furthermore all nodes have been used to initiate the shortest paths search, and homopolar bonds have been taken into account in the analysis.

Computations have been performed using a code developed to implement our method and named Rigorous Investigation of Networks Generated using Simulations, 'RINGS', see Appendix C for further details.

4.2.2. Results

On the one hand we have decided to evaluate the standard $R_N(n)$ value [Section 3.1], and on the other hand we have compared this information with the connectivity profiles obtained using our method which includes $R_C(n)$, $P_N(n)$, $P_{\max}(n)$ and $P_{\min}(n)$.

$R_N(n)$ is presented in Fig. 13 whereas the connectivity profiles of SiO_2 and GeS_2 glasses are presented in Fig. 14.

First of all we can see that the SiO_2 and the GeS_2 glassy networks are really different, this observation was predictable considering the numerous studies already published on these two systems [7,12–14].

As already proposed it appears that bond defects have a clear impact on the ring statistics of $g-GeS_2$. Thus rings with homopolar bonds, non-even sized rings in total number of nodes, exist in all the range of sizes (5–29 nodes) in $g-GeS_2$ [Figs. 13, 14]. This observation confirms the influence of these bond defects on the connectivity of the network.

Moreover we obtain similar information with $R_N(n)$ [Fig. 13] and $R_C(n)$ [Fig. 14a]. Indeed $R_N(n)$ and $R_C(n)$ highlight almost the same properties for the SiO_2 network (importance of the rings of size 8–14 nodes) as well as for the GeS_2 network (importance of the rings of size 4, 6 and 24–29 nodes).

Nevertheless the analysis of the connectivity profiles [Fig. 14] reveals much more information. On the one hand $R_C(n)$ [Fig. 14a] emphasizes ring sizes in significant proportions, on the other hand $P_N(n)$ [Fig. 14b] emphasizes statistical effects on $R_C(n)$. As previously illustrated in Fig. 11 a ring is a solution of the shortest path algorithm [Section 2.2.2] for at least one particular node but not necessarily for any other node in this ring (in Fig. 11 the rings with 10 nodes are not solutions of the search for all the nodes that constitute them). A high proportion of rings with n nodes, $R_C(n)$, might emphasize a characteristic of the network, nevertheless this characteristic would be considered global or significant only if these rings are solutions of the search for a high proportion of nodes, $P_N(n)$. Thus it appears that $R_C(n)$ [Fig. 14a] and $P_N(n)$ [Fig. 14b] bring complementary information on the proportions of rings in the two materials.

A high value of $R_C(n)$ and a low value of $P_N(n)$ will therefore simply illustrate the existence of a local network defect. This is

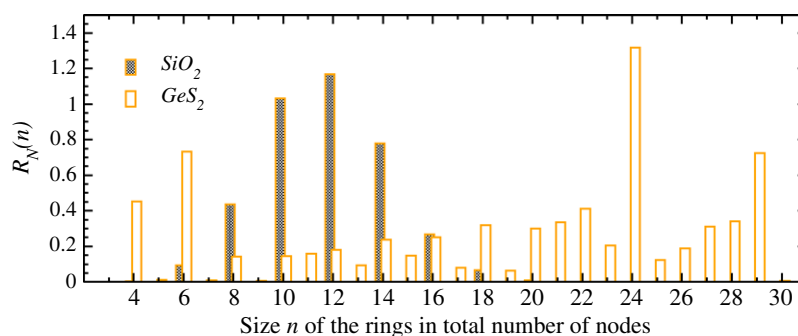


Fig. 13. Rings per node $R_N(n)$ of SiO_2 and GeS_2 glassy samples.

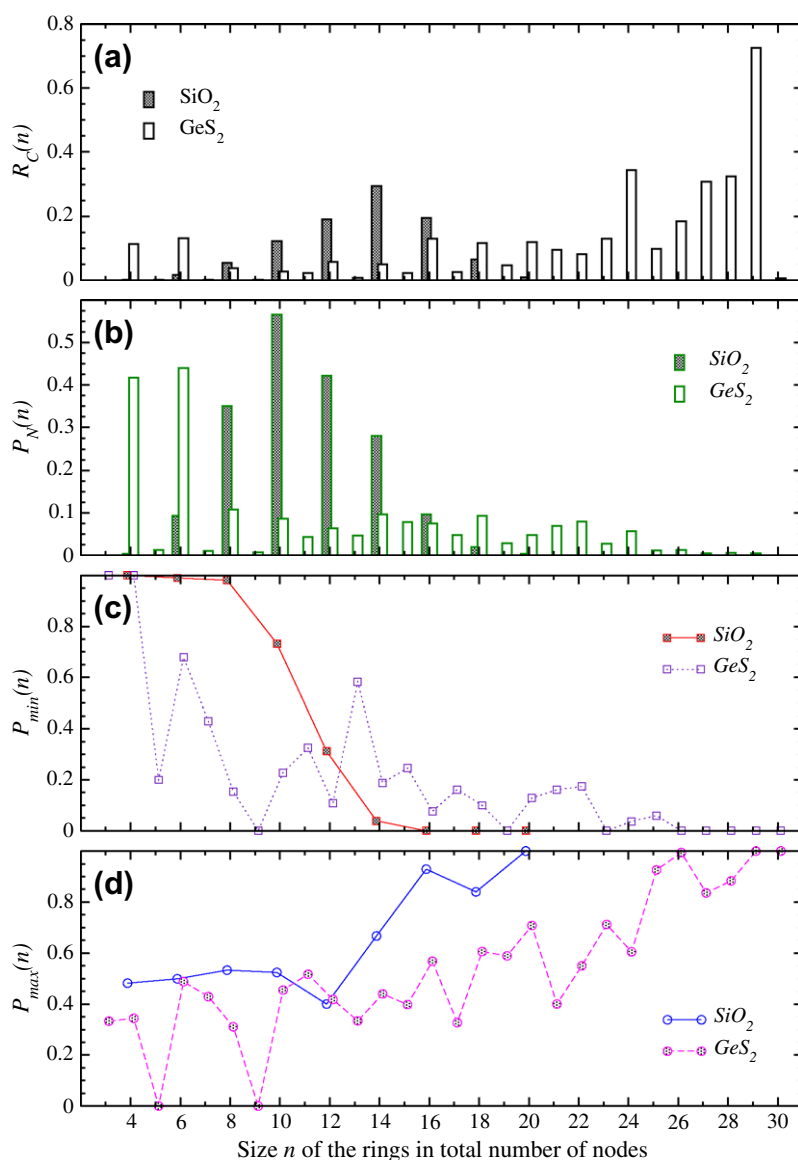


Fig. 14. Connectivity profiles of SiO_2 and GeS_2 glassy samples.

illustrated in Fig. 14a and b by the rings with 16 nodes in SiO_2 and even more clearly by the rings with 24–29 nodes in GeS_2 . On a general point of view the larger the size n of the shortest path for a particular node, the more probable it is to find different rings of the same size n and solutions as well of the shortest path algorithm for this node. This is true in particular in GeS_2 with the presence

of Edge-Sharing tetrahedra that can literally double the number of rings when located on the path (see Fig. 3-1 for an idea).

At the opposite, a high value of $P_N(n)$ will highlight a global characteristic of the network, for example a sub-network of rings with n nodes. This is illustrated in Fig. 14a and b by the rings with 10 and 12 nodes in SiO_2 and the rings with four and six nodes in

Table A.6

General connectivity matrix for the networks represented in Fig. 9 and studied using the different definitions of rings.

	King/S.P.	Primitive/strong
Case (a)	$\begin{bmatrix} 4/16 & 2/16 \\ 2/16 & 5/16 \end{bmatrix}$	$\begin{bmatrix} 4/16 & 4/16 \\ 4/16 & 7/16 \end{bmatrix}$
<i>All criteria</i>		
Case (b)	$\begin{bmatrix} 4/16 & 0/16 \\ 0/16 & 12/16 \end{bmatrix}$	
Case (c)	$\begin{bmatrix} 4/16 & 1/16 \\ 1/16 & 12/16 \end{bmatrix}$	
Case (d)	$\begin{bmatrix} 4/16 & 0/16 \\ 0/16 & 11/16 \end{bmatrix}$	
Case (e) → (f)	$\begin{bmatrix} 4/16 & 2/16 \\ 2/16 & 12/16 \end{bmatrix}$	
Case (g) → (i)	$\begin{bmatrix} 4/16 & 1/16 \\ 1/16 & 11/16 \end{bmatrix}$	
n – ring with n nodes	$\begin{bmatrix} n4 & n6/n4 \\ n4/n6 & n6 \end{bmatrix}$	

Table A.7

$P_N(n)$ – proportion of nodes at the origin of at least one ring of size n for the networks presented in Fig. 9.

n	$P_N(n)$	
	King/S.P.	Primitive/strong
Case (a)		
4	4/16	4/16
6	5/16	7/16
n	$P_N(n)$	
<i>All criteria</i>		
Case (b) → (c)		
4	4/16	
6	12/16	
Case (d)		
4	4/16	
6	11/16	
Case (e) → (f)		
4	4/16	
6	12/16	
Case (g) → (i)		
4	4/16	
6	11/16	

Table A.8

Connectivity profiles results of the ring statistics for the networks presented in Fig. 9.

n	$R_C(n)$	$P_N(n)$	$P_{\max}(n)$	$P_{\min}(n)$
Case (a)	King/S.P.			
4	1/16	4/16	0.5	1.0
6	2/16	5/16	1.0	0.6
Case (a)	Primitive/Strong			
4	1/16	4/16	0.5	1.0
6	2/16	7/16	1.0	3/7
Case (b)	All criteria			
4	1/16	4/16	1.0	1.0
6	2/16	12/16	1.0	1.0
Case (c)				
4	1/16	4/16	0.75	1.0
6	2/16	12/16	1.0	11/12
Case (d)				
4	1/16	4/16	1.0	1.0
6	2/16	11/16	1.0	1.0
Case (e) → (f)				
4	1/16	4/16	0.5	1.0
6	2/16	12/16	1.0	10/12
Case (g) → (i)				
4	1/16	4/16	0.75	1.0
6	2/16	11/16	1.0	10/11

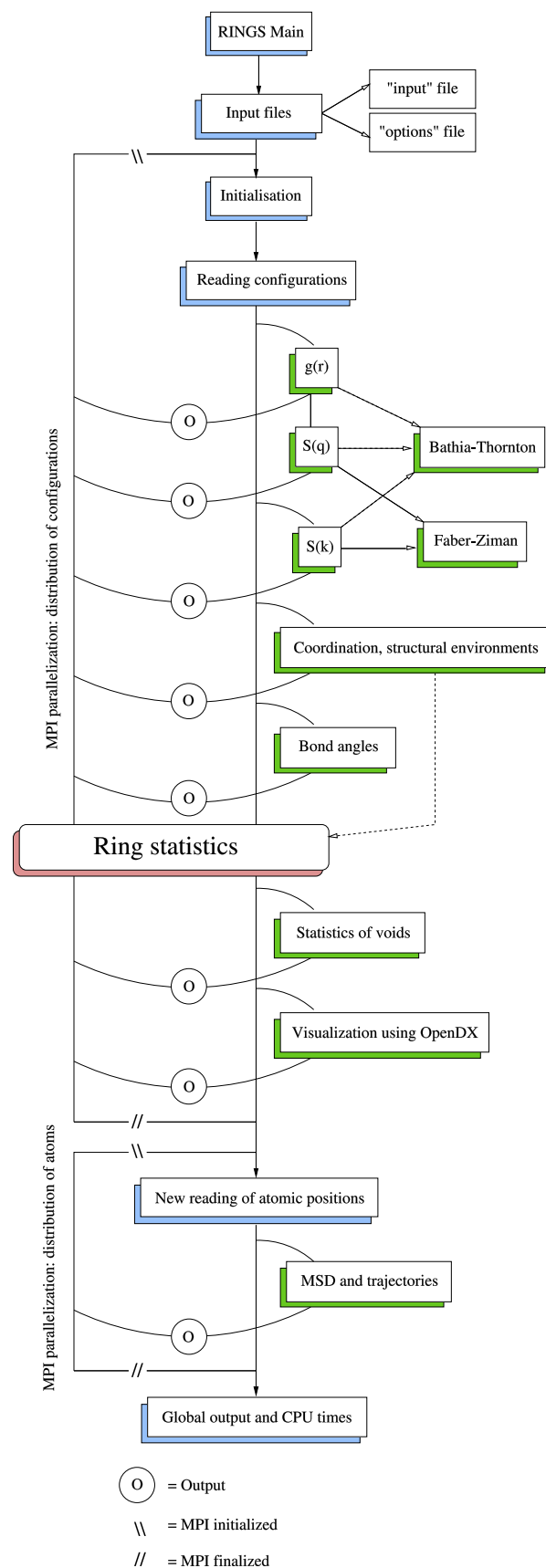


Fig. C.15. Simplified representation of the RINGS code structure.

GeS₂. Furthermore it has to be noticed that this information can not be obtained using only the quantities $R_C(n)$ and/or $P_N(n)$.

Before any discussion regarding the values of $P_{\max}(n)$ [Fig. 14c] and $P_{\min}(n)$ [Fig. 14d] it is interesting to remind that these values, like the $R_C(n)$ and $P_N(n)$ values, refer to a global picture of the connectivity. Thus, for GeS_2 as for SiO_2 , $P_{\max}(n)$ and $P_{\min}(n)$ refer to averages of the properties of the *network forming* nodes (Si in SiO_2 and Ge in GeS_2) and the properties of the *network modifying* nodes (O in SiO_2 and S in GeS_2). Using the approximation presented in Appendix B it is possible to estimate the contributions of each chemical species to $P_{\min}(n)$ and $P_{\max}(n)$.

Considering $P_{\max}(n)$ [Fig. 14c] and $P_{\min}(n)$ [Fig. 14d] we can obtain information on the distribution of the rings in SiO_2 and GeS_2 :

- In g- SiO_2 it appears for $n \leq 8$ that $P_{\min}(n) \simeq 1$ [Fig. 14c]. Therefore rings with $n \leq 8$ nodes are always the shortest paths for the node at the origin of the search. Thus one particular node can not be at the origin of rings with four and/or six and/or eight

nodes. In other words in g- SiO_2 there is no common node between rings with four, six and eight nodes. For $n > 8$, $P_{\min}(n)$ decreases until being equal to 0 for $n = 16$ [Fig. 14c], then smaller rings exist.

Furthermore in g- SiO_2 it appears for $n \leq 12$ that $P_{\max}(n) \simeq 0.5$ [Fig. 14d]. Using the approximation [Eq. (B.2)] it is possible to estimate for $n \leq 12$ that $P_{\max}(n)[\text{Si}] \sim 0$. Which means that for $n \leq 12$ when a ring is found, this ring is never the largest ring that can be found. Thus there is no Si atom having a connectivity with only rings with four, six, or eight nodes. Then for $n \text{ GeS}_2$, $P_{\max}(n) \simeq 1$ [Fig. 14d] and very few larger rings exist. In other words in g- SiO_2 a very small number of the rings with 16, 18 and 20 nodes have a node in common.

Then it is possible to view the ring network of g- SiO_2 as an homogeneous alternating of rings with n nodes, rings with m nodes, and rings with o nodes, with $n \in [4-8]$, $m \in [10-12]$ and

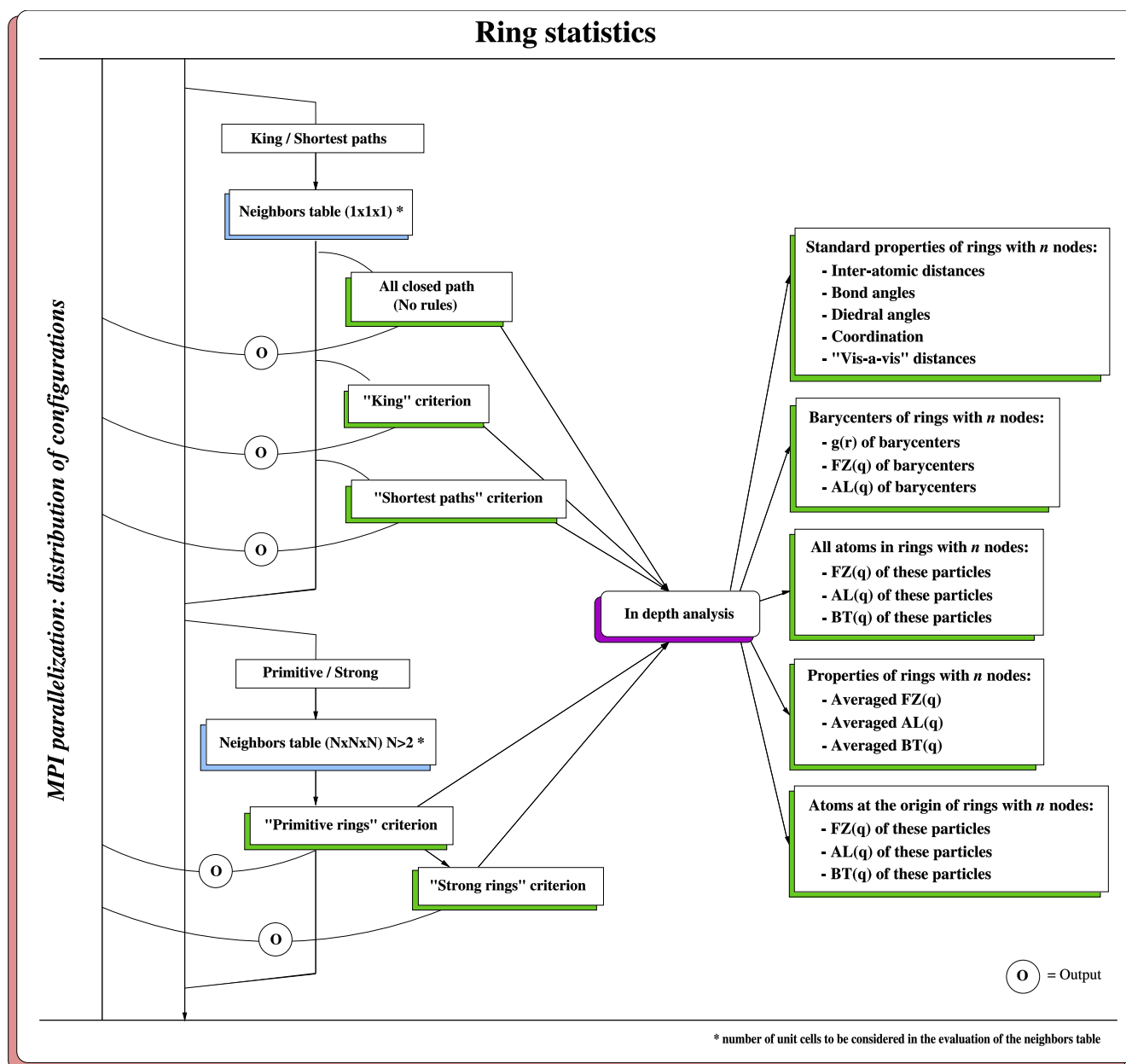


Fig. C.16. Ring statistics in the RINGS code – simplified structure

$o \in [14-20]$ nodes. In this construction the rings are connected together by the Si atoms.

- In g-GeS₂ the disorder is much more important. This is illustrated in particular by the peaks in $P_{\min}(n)$ [Fig. 14c] and $P_{\max}(n)$ [Fig. 14d] for the sizes of rings with homopolar bonds. Thus for rings with five and nine nodes low values of both $P_{\min}(n)$ and $P_{\max}(n)$ illustrate the existence of local defects with no real impact on the connectivity. Furthermore for rings with three, 11, and 13 nodes, significant values of $P_{\min}(n)$ and/or $P_{\max}(n)$ illustrate the existence of a defect with a serious impact on the connectivity. The evolutions of both $P_{\min}(n)$ and $P_{\max}(n)$ are much more complicated than the ones observed for SiO₂.

For $n = 6$ $P_{\min}(n) = 0.7$, using the approximation [Eq. (B.2)] it is possible to estimate for $n = 6$ that $P_{\max}(n)[\text{Ge}] \sim 0.4$. Thus in g-GeS₂ there is a significant proportion of intersections between the rings with four and six nodes. Furthermore for $n \in [12-22]$ $P_{\min}(n) \neq 0$, thus even rings with 12–22 nodes can be the shortest solutions of the search for a particular node. This is confirmed by the $P_{\max}(n)$ values in the range $n \in [12-22]$ [Fig. 14d]. Indeed for $n \in [12-22]$, $P_{\max}(n) \sim 0.5-0.7$, which highlights the existence of zones of very large rings, being shortest paths [Section 2.2.2], in GeS₂.

This highlights once again the defectuous nature of the chalcogenide network. In g-GeS₂ the rings distribution appears to be very inhomogeneous with zones of small rings (4–10 nodes) and zones of large, up to very large rings (≥ 12 nodes).

5. Conclusion

First of all we have presented the state of the art of ring statistics. Then we have emphasized that the standard tools used to study the connectivity of materials using these techniques were not suitable. Therefore we have proposed a new and original approach to analyse the connectivity of materials using ring statistics. We have highlighted that our method allows to obtain the connectivity profile of a material. This profile includes the number of rings of size n nodes in the material, $R_c(n)$, the proportion of nodes for which looking for rings returns at least one ring of size n , $R_N(n)$, and the probabilities $P_{\max}(n)$ and $P_{\min}(n)$, that a ring with n nodes represents respectively the longest or the shortest ring when found during the search.

Afterwards we have illustrated the features of our method by studying the connectivity of amorphous g-SiO₂ and g-GeS₂ systems. Then we have shown that the information contained in $R_c(n)$ and $P_N(n)$ were complementary and allow to obtain an accurate description of the proportions of rings in the analysed networks. Furthermore we have shown that $P_{\max}(n)$ and $P_{\min}(n)$ can be used to distinguish the distribution of rings in the analysed networks, and to obtain information on the global organisation of these networks.

Moreover almost all the information provided by the connectivity profile is inaccessible using the standard tools of the ring statistics. Thus this method allows to obtain many new aspects of the analysis of a network material.

With the example of SiO₂ and GeS₂ glasses we have seen that our method allows to study and compare with efficiency two amorphous systems. It appears now really interesting to apply this method to a wide variety of amorphous systems and to classify glasses based upon their connectivity profiles and to compare this classification with their standard physical properties.

Appendix A. Tables related to the discussion of Section 3.2.2

Tables A.6–A.8

Appendix B. Decomposition and approximation of $P_{\max}(n)$ and $P_{\min}(n)$ in terms of the partials contributions of the chemical species for an AB₂ system

By decomposing $P_{\min}(n)$ or $P_{\max}(n)$ in terms of the partials contributions of the chemical species for an AB₂ system, we find:

$$P_m(n) = \frac{N_m(n)[A] + N_m(n)[B]}{N(n)[A] + N(n)[B]} \quad (\text{B.1})$$

with $N(n)[X]$ the number of atoms of the chemical species X where the search returns rings with n nodes, $N_m(n)[X]$ the number of atoms of the chemical species X where rings with n nodes are the shortest or the longest paths ($m = \min, \max$) among the different solutions proposed by the analysis. The coordination of the chemical species are quite similar in SiO₂ and GeS₂ with $N_o(\text{Si}) \simeq N_s(\text{Ge}) \simeq 4$ for the *network forming* nodes A and $N_{\text{Si}}(\text{O}) \simeq N_{\text{Ge}}(\text{S}) \simeq 2$ for the *network modifying* nodes B. On a logical point of view only 1

```
#####
#                               #
#          RINGS Input file     #
#####
(a) GeS2-glass                  #
(b) 258                        #
(c) 2                          #
(d) Ge S                        #
(e) 1000                       #
(f) 1                          #
(g) 19.209999 0 0              #
    0 19.209999 0              #
    0 0 19.209999              #
(h) 2.5                        #
(i) ANI                        #
    ges2.xyz                   #
(j) 200                        #
(k) 500                        #
(l) 25                         #
(m) 0.125                      #
(n) 90                         #
(o) 10                         #
(p) 15                         #
#####
(q) Ge Ge 4.5                  #
    S S 4.70                   #
    Ge S 2.3                   #
(r) Gtot 2.60                  #
#####
```

- (a) The name of the system studied.
- (b) The number of atoms.
- (c) The number of chemical species.
- (d) The list of chemical species.
- (e) The number of configurations.
- (f) The format of the box parameters.
- (g) The box edges parameters.
- (h) Integration time step Δt of the Newton's equations of motion.
- (i) File format for the configurations:
The next line depends on the file format which can be:
- Multiple xyz (ANI)
- Fireball msf (MSF)
- CPMD trajectory (TRJ)
- VASP trajectory (VAS)
- (j) Real space discretization for the $g(r)$ calculations.
- (k) Reciprocal space discretization for the $S(q)$ calculations.
- (l) Maximum modulus of the reciprocal space vectors for the $S(q)$ calculations.
- (m) Smoothing factor for the $S(q)$ computed using the Debye equation.
- (n) Angular discretization.
- (o) Real space discretization for the void and the ring statistics.
- (p) Maximum search depth for the ring statistics.
- (q) Cutoff radius from the partials $g(r)$ (optionals).
- (r) Total cutoff radius from the total $g(r)$ (optional).

Fig. C.17. Sample 'input' file for the RINGS code.

```
#####
RINGS options file
#####
① PBC .true. #
② Frac .false. #
③ g(r) .false. #
④ S(q) .false. #
⑤ S(k) .false. #
⑥ MSD .false. #
⑦ atMSD .false. #
⑧ Bonds .true. #
⑨ Angles .false. #
⑩ Rings .true. #
---- ! Ring statistics options ! ----
⑪ Species 0 #
⑫ ABAB .false. #
⑬ Rings0 .false. #
⑭ Rings1 .false. #
⑮ Rings2 .false. #
⑯ Rings3 .true. #
⑰ Rings4 .true. #
⑱ Prim_Rings .true. #
⑲ Str_Rings .false. #
⑳ BarycRings .false. #
㉑ Prop-1 .false. #
㉒ Prop-2 .false. #
㉓ Prop-3 .false. #
㉔ Prop-4 .false. #
㉕ Prop-5 .false. #
-----
㉖ Voids .false. #
#####
Output options
#####
㉗ Evol .false. #
㉘ Dxout .false. #
#----! Check OpenDX plot options ! ---#
㉙ RadOut .false. #
㉚ RingsOut .false. #
㉛ DRngOut .false. #
㉜ VoidsOut .false. #
㉝ TetraOut .false. #
㉞ TrajOut .false. #
-----
㉟ Output rings.out #
#####
```

- ① Use or not the periodic boundary conditions.
- ② Fractional (.true.) or real (.false.) atomic coordinates.
- ③ Radial distribution functions, $g(r)$:
 - Total $g(r)$ and total $G(r)$.
 - Partial $g_{\alpha\beta}(r)$ and partial $G_{\alpha\beta}(r)$.
 - Bathia-Thornton $g(r)$ [17].
- ④ Neutron and X-rays structure factors, $S(q)$, from FFT of the $g(r)$ (depends on ③):
 - Various neutron and X-rays weighted functions ($G(r)$, $T(r)$, $D(r)$, $g(r)$) [18].
 - Total $S(q)$.
 - Faber-Ziman partials $S_{\alpha\beta}(q)$ [19].
 - Ashcroft-Langreth partials $S_{\alpha\beta}(q)$ [20, 21, 22].
 - Bathia-Thornton partials $S(q)$ [23].
- ⑤ Neutron and X-rays structure factors, $S(q)$ using the Debye equation:
 - Total $S(q)$.
 - Faber-Ziman partials $S_{\alpha\beta}(r)(q)$.
 - Ashcroft-Langreth partials $S_{\alpha\beta}(q)$.
 - Bathia-Thornton partials $S(q)$.
- ⑥ Mean Square Displacement, MSD.
 - MSD.
 - Uncorrelated MSD.
 - Directional MSD (x , y , z , xy , xz , yz).
 - Directional uncorrelated MSD (x , y , z , xy , xz , yz).
 - Drift of the center of mass.
 - Diffusion constant.
- ⑦ MSD for each single atom.
- ⑧ Properties of the chemical bonds:
 - Coordination.
 - Proportions of the different structural environments.
 - Proportions of tetrahedra.
 - Proportions of inter-tetrahedral bonds.
- ⑨ Bond angles and dihedral angles distribution.
- ⑩ Activation of the ring statistics.
 - ⑪ Chemical species used to initiate the search (depends on ⑧):
 - 0 = All atoms are used
 - 1 = Only the species of highest coordination is used (network forming nodes).
 - 2 = Only the species of lowest coordination is used.
 - ⑫ Only look for ABAB rings.
 - Homopolar bonds are ignored when creating the nearest neighbors table.
 - Allows to look for primitive ABAB rings.
 - Look for ABAB rings in systems with more than 2 chemical species.
 - The option affects all the RINGS code as well as options ⑯ and ⑰.
 - ⑬ Search for all closed paths in the box.
 - ⑭ Search for shortest paths without homopolar bond(s).
 - ⑮ Search for King's rings without homopolar bond(s).
 - ⑯ Search for shortest paths with homopolar bond(s).
 - ⑰ Search for King's rings with homopolar bond(s).
 - ⑱ Search for primitive rings.
 - ⑲ Search for strong rings.
 - ⑳ Trajectories of the barycenters of the rings.
 - Create an input file for visualization using OpenDX.
 - ㉑ Advanced properties 1:
 - Inter-atomic distances in rings with n nodes.
 - Bond angles in rings with n nodes.
 - Coordination numbers in rings with n nodes.
 - "Vis-a-vis" distances in rings with n nodes.
 - ㉒ Advanced properties 2:
 - Radial distribution function, $g_n(r)$, of the barycenters of rings with n nodes.
 - rings with n nodes barycenters neutron structure factor *.
 - ㉓ Advanced properties 3:
 - Neutron structure factor of particles at the origin of rings with n nodes *.
 - ㉔ Advanced properties 4:
 - Neutron structure factor of all particles involved in rings with n nodes *.
 - ㉕ Advanced properties 5:
 - Average neutron structure factor of rings with n nodes *.

* Total and partials in the F.-Z. [19], A.-L. [20, 21, 22] and B.-T. [23] formalisms.

- ㉖ Statistics of voids.
- ㉗ Output of $g(r)$, $s(q)$, ... for each configuration.
- ㉘ OpenDX visualization file.
 - ㉙ Use atomic radius given in the 'input' file as color map.
 - ㉚ Use the connectivity factor F_c [18] as color map.
 - ㉛ Use ΔF_c [18] as color map.
 - ㉜ Add free volumes in the OpenDX visualization file.
 - ㉝ Add tetrahedral environments in the OpenDX visualization file.
- ㉜ Visualization of the trajectories of the particles of highest MSD using OpenDX.
- ㉞ Name of the global output file.

Fig. C.18. Sample 'options' file for the RINGS code. (See above-mentioned references for further information.)

size of ring [Eq. (3)] can be found per *network modifying* node (O or S). Therefore $N_m(n)[B] \simeq N(n)[B]$ and for each size of ring $P_{\min}(n)[B] \simeq P_{\max}(n)[B]$ where $P_{\min}(n)[B]$ and $P_{\max}(n)[B]$ represent the partial contributions of the *network modifying* nodes. Further-

more considering the proportion of each chemical species, in an AB_2 system forming mostly ABAB rings it is legitimate to postulate that $N(n)[B] \sim N(n)[A]$. Following these remarks the equation [Eq. (B.1)] can be approximated by:

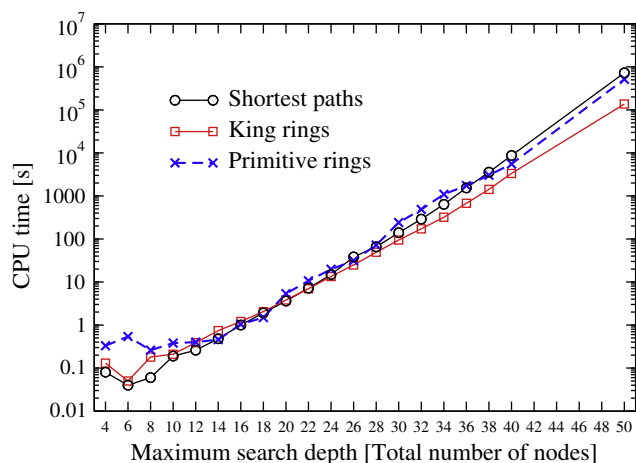


Fig. C.19. Evolution of the CPU time as a function of the maximum search depth. A single configuration of g-GeS₂ at 300 K (258 particles) has been analyzed. Furthermore all nodes have been used to initiate the search. (3600 s = 1 h, 86,400 s = 24 h, 864,000 s = 10 days, 2,592,000 s = 1 month). Computations were performed using an Intel Xeon 2.6 GHz processor with 1.5 Go of memory.

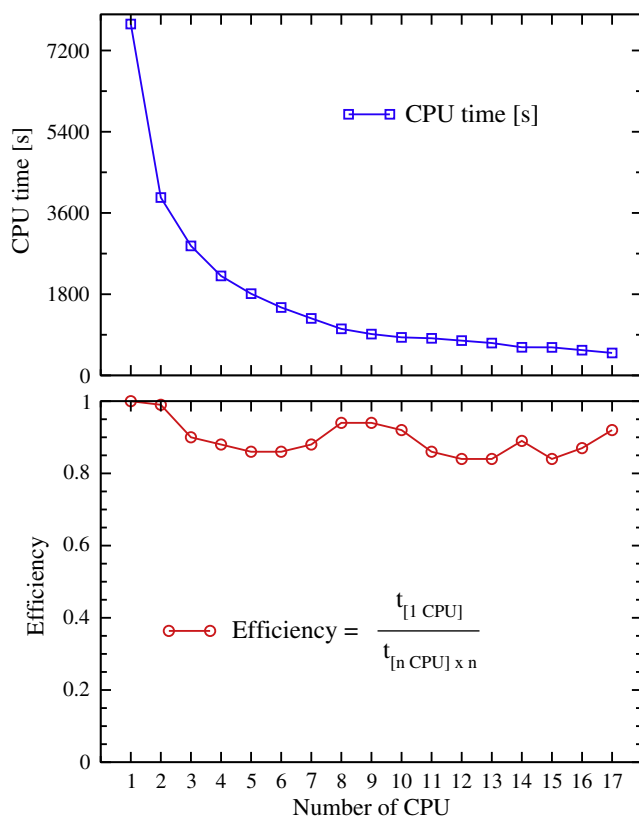


Fig. C.20. Evaluation of the parallelization in the RINGS code. Evolution of the CPU time and the efficiency as a function of the number of CPU. 100 configurations of g-GeS₂ (258 particles) at 300 K have been analysed using the shortest path criterion. All nodes have been used to initiate the search and the maximum search depth has been set to 30 (in total number of nodes). Computations were performed using an IBM blade center using Intel Xeon 2.6 GHz processors with 1.5 Go of memory.

Appendix C. The RINGS code

In order to analyse the results of our molecular dynamics simulations we have developed a code named Rigorous Investigation of Networks Generated using Simulations, 'RINGS'. The RINGS code has been written in FORTRAN 90 and is almost 30,000 lines of instructions long (with 210 subroutines and four modules). The memory allocation and deallocation processes are mostly dynamical. Furthermore the code is fully parallelized using the Message Passing Interface, 'MPI' [16] with an unblocking scheme.

The RINGS code is mainly devoted to the ring statistics nevertheless many other analysis can be performed. The many features of the code are illustrated in Fig. C.15 which represents a simplified diagram of its structure. It has to be noticed that liquid, amorphous, as well as crystalline systems can be studied using the RINGS code.

The part of the RINGS code dedicated to the ring statistics is presented in Fig. C.16. The analysis regarding the different types of rings introduced in the first part of the article [Section 2.2] have been implemented (King, shortest paths, primitive and strong rings). During the analysis the standard values (R_N and R_C), the connectivity matrix [Table 5] and the connectivity profiles as defined in our method ($R_C, P_N, P_{max}, P_{min}$) are computed. Furthermore Fig. C.16 illustrates that the RINGS code does not focus only on the connectivity, i.e. finding the rings, but also on the physical properties which can be deduced from the analysis. Further studies have been scheduled to use these features of the code.

As shown in Fig. C.15 two files are needed to setup a calculation using the RINGS code, an 'input' file [Fig. C.17] which describes the system, as well as an 'options' file [Fig. C.18] which describes the calculations to be performed.

As shown in Fig. C.18 numerous analysis can be performed using the RINGS code. Nevertheless the CPU time is mainly driven by the ring statistics analysis, in particular by the maximum search depth, i.e. the maximum size for a ring which is an input parameter [Fig. C.17] fixed during the analysis. This has been illustrated in Fig. C.19 which represents the evolution of the CPU time as a function of the maximum search depth. As shown in Fig. C.19 the CPU times increase exponentially with the maximum search depth.

Furthermore we can see in Fig. C.19 that the evolution of the CPU time is similar in the cases of King, shortest path and primitive rings criteria (the CPU times for the strong ring criterion, not shown here, are much more important). Therefore the MPI parallelization appears fully justified to performed the ring statistics analysis of network material with a high search depth.

The results of the parallelization are illustrated in Fig. C.20 which represents the evolution of the CPU time as well as of the efficiency of the MPI modifications, as a function of the number of CPU used in the calculation. The unblocking scheme which has been chosen for the parallelization appears to be really efficient [Fig. C.20]. However such a behaviour could be expected when distributing the configurations on the processors.

In addition to the features of the RINGS code which have already been presented, visualization modules have been developed using <http://www.opendx.org> OpenDX, the open source software version of IBM's Visualization Data Explorer Product. The RINGS code allows indeed to produce input files for several OpenDX tools, including a visualization interface for molecular dynamics which allows to export pictures as well as animations of the simulation. Furthermore it is possible to add free volumes, coordination tetrahedra or other information to the visualization window. An example is presented in Fig. C.21.

The RINGS code is protected by the terms of the CeCILL-B license [24]. The RINGS code as well as the OpenDX modules are available at <http://rings-code.sourceforge.net>.

$$P_m(n) \sim \frac{N_m(n)[A] + N(n)[B]}{N(n)[A] + N(n)[B]} \sim \frac{P_m(n)[A] + 1}{2} \quad (\text{B.2})$$

where $P_m(n)[A]$ represents the partial contribution of the network forming nodes to $P_m(n)$.

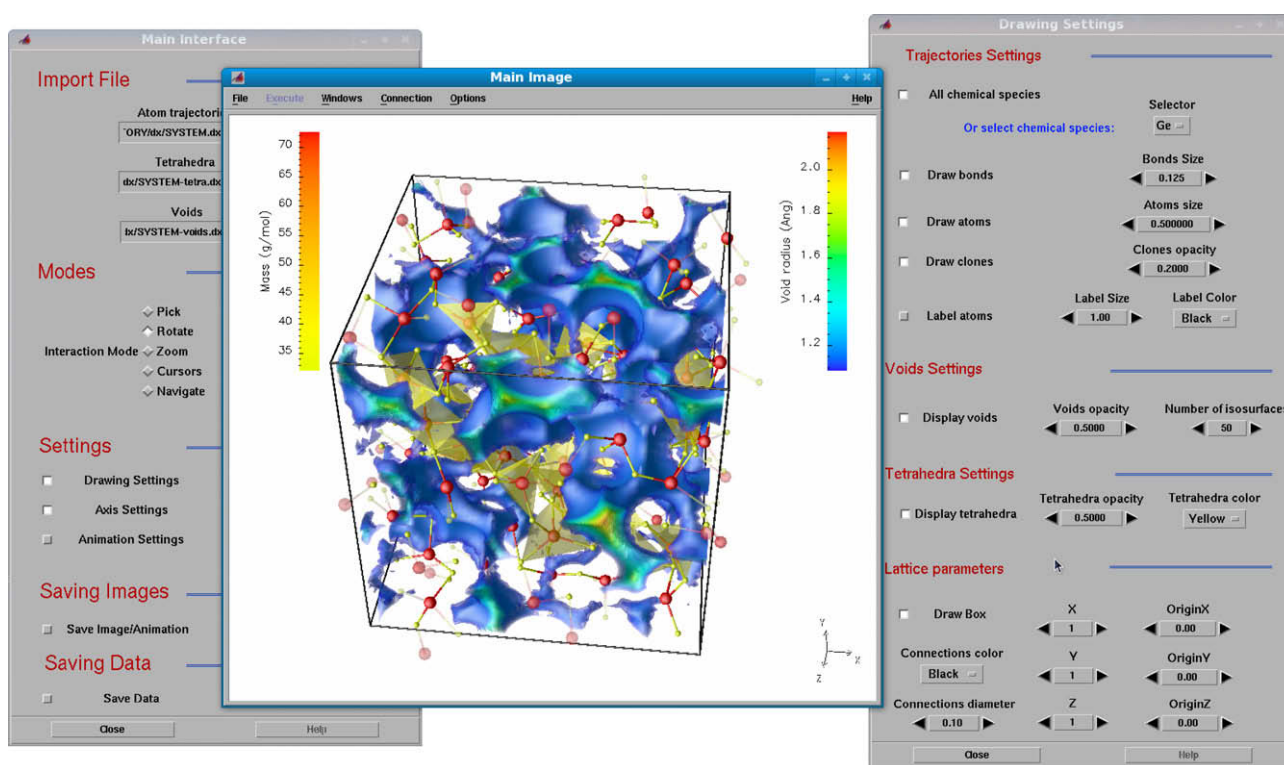


Fig. C.21. OpenDX visualization interface for the RINGS code.

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Erratum

Erratum: Ring statistics analysis of topological networks: New approach and application to amorphous GeS₂ and SiO₂ systems. [Comput. Mater. Sci. 49 (2010) 70–83]

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- The first definition of a ring presented in our article, erroneously attributed to King [1] was actually provided by L. Guttman [2]. Thus Guttman's name should be associated to the definition presented in Section 2.2.1 and King's name to the definition in Section 2.2.2
- Our expression of the theoretical maximum size of a ring that can be found in a network, TMS, for the second definition of a ring presented in the article (Section 2.2.2) has to be corrected:

$$TMS = 2 \times (D_{max} - 2) \times (N_{cmax} - 2) + 2 \times D_{max} \quad (1)$$

where D_{max} is the longest distance, in terms of chemical bonds, separating two atoms in the network, and N_{cmax} represents the average number of neighbors of the chemical species of highest coordination.

- The size of the general connectivity matrix presented in Section 3.2.2 is $(R - r + 1) \times (R - r + 1)$, the size of this matrix can change depending on the number of ring sizes that can be found in the network. Furthermore the matrix can be reshaped as we did it in the article when presenting in table A.6 the connectivity matrix of the networks in Fig. 9: two sizes of ring exist for the networks in Fig. 9, thus the connectivity matrix has a dimension of 2×2 .

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